

Royalty Calculation App Documentation

Summary

The Royalty Calculation app is a Retool-based financial management tool designed to track sales, calculate royalty adjustments, manage Minimum Guarantee (MG) balances, and generate monthly royalty reports. It provides a workflow for closing monthly royalty periods and managing contract transfers between licensors.

Main Purpose and Workflow

The application serves as a centralized interface for:

1. **Sales Management:** Viewing, creating, and updating sales records.
2. **Royalty Calculation:** Calculating royalties based on sales and adjustments, including tracking MG balances.
3. **Monthly Closing:** Finalizing monthly royalty data and locking periods to prevent further modifications.
4. **Reporting:** Generating Excel-based royalty reports for specific licensors and contracts.

Major Data Flows:

- **Input:** Users provide date ranges and select filters (Licensor, Game Title, GL Account) via UI widgets.
- **Processing:** Queries fetch data from the `NISA_mysql_connection`. JavaScript logic (e.g., `calculateMGBalance`, `exportReport`) processes this data to compute balances and format Excel files.
- **Output:** The app displays data in tables and allows users to export reports or update records in the database.

Technical Components

Key SQL Queries

- `get_all_items`: Retrieves transaction data for the selected date range.
- `calc_current_mg_balance`: A complex query using `UNION` to calculate MG balances by comparing `v_all_transactions` and `v_all_royalty_adjustment` against contract terms.
- `create_changed_records` / `create_open_original_records`: Used to generate billing records based on whether the month is closed or open.
- `insert_monthly_royalty`: Performs a bulk insert into the `monthly_royalty` table.

JavaScript Logic

- `calculateMGBalance`: Orchestrates the calculation of MG balances by triggering multiple sub-queries (`get_royalty_due_from_closed_month`, `get_royalty_due_from_open_month`, etc.) and aggregating the results.
- `exportReport`: Uses the `ExcelJS` library to generate and download multi-sheet Excel reports. It iterates through months, fetches data, applies formatting (borders, number formats), and triggers a

file download.

- `transfer_licensors`: A recursive function that handles the transfer of contracts between licensors by updating the `active_contract` table and inserting new records into the `contract` table.

Key Widgets

- `tabbedContainer1`: The primary navigation component separating the app into functional areas (Sales, Royalty Calculation, Adjustment, Billing, Report, Close Month).
- `royalty_table`: Displays transaction data with conditional row coloring based on unit price changes.
- `delete_modal`: A safety-gated modal for deleting records, which checks if a period is closed before allowing deletion.

Inferences

- **Data Integrity:** It is inferred that the `version_id` column in various tables (e.g., `royalty_adjustment`, `sales`) is used for optimistic locking or audit history, as the app frequently increments this value during updates.
- **Database Schema:** Based on the queries, the database appears to rely on views (e.g., `v_all_transactions`, `v_all_royalty_adjustment`) to abstract complex joins between `licensor`, `contract`, `game_title`, and `sales` tables.

Appendix: Inventory

Major Queries

Query Name	Responsibility
<code>get_all_items</code>	Fetches primary transaction data for the main table.
<code>calc_current_mg_balance</code>	Calculates MG balances for contracts.
<code>insert_monthly_royalty</code>	Persists calculated royalty data to the database.
<code>get_sales</code>	Retrieves sales data for the Sales tab.

update_royalty_adjustment	Updates existing adjustment records.
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JavaScript Logic Blocks

Block Name	Role
calculateMGBalance	Aggregates royalty and adjustment data to compute MG status.
exportReport	Generates and downloads Excel reports using ExcelJS.
transfer_licensors	Manages contract migration between licensors.
delete_record_transform	Validates and executes record deletion.

Important Widgets

Widget Name	Interaction
tabbedContainer1	Main navigation between functional modules.
royalty_table	Main data display; supports row selection and editing.
delete_modal	Safety gate for record deletion.
start_date / end_date	Global date filters for data retrieval.

Risks and Open Questions

- Performance:** The calc_current_mg_balance query is highly complex with multiple subqueries. As the dataset grows, this may lead to timeouts or performance degradation.
- Data Consistency:** The app relies on isClosedMonthStatus to determine logic paths. If the status in monthly_royalty_status is not updated correctly, the calculation logic may produce incorrect results.
- Dependency:** The exportReport function relies on an external library (ExcelJS). If the CDN link is broken or the library API changes, report generation will fail.

- **Hardcoded Logic:** Several queries contain hardcoded logic (e.g., `systemStartDate = '2022-07-01'`), which may require manual updates in the future.